Survey Research and Design in Psychology

Tutorial 5 - Public data analysis

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Overview

- Apply techniques learnt in this course to real world data
- Boolean logic

Data

- Data is constantly being collected. Some examples include:
  - Research
    - Surveys
    - Customer satisfaction
    - Marketing or polls
  - Phone/Internet usage
    - Applications
    - Streaming services (e.g. YouTube, Netflix etc.)
  - Biometrics (e.g. smart watches)
    - Sleep
    - Heart rate
    - Physical activity
  - Location
  - Healthcare
- Data is beautiful thread on reddit

Data

- Important considerations include:
  - Privacy
  - Digital footprint
  - Public access
    - Australian government
    - GitHub - awesome-public-datasets
    - PsyArXiv, medRxiv and bioRxiv
    - Gapminder

Gapminder data

- Open Gapminder data

Question

- How long are people living all around the world according to this data set?
  - Generate descriptive statistics and a histogram

Question

- Does worldwide life expectancy change across the years in the data set? i.e. are people living longer as we progress through time?
  - What is the dependent variable? What is the independent variable?
  - Generate a scatterplot - think about what should go on the x and y axis?

Question

- How does the relationship between year and worldwide life expectancy vary by continent?
  - Generate a scatterplot

Question

- What is the life expectancies of Australians?
  - Hint: use filters, then plot the scatterplot as before

Question

- How do the trends in life expectancy for Australians compare with those living in Ireland?
  - We need boolean logic!

Boolean logic

- Boolean logic is a type of algebra in which results are calculated as either TRUE or FALSE (known as truth values or truth variables)

Boolean operators

- AND
- OR
- NOT

Boolean logic - AND

A	B	A&B
1	1	1
1	0	0
0	1	0
0	0	0

Boolean logic - OR

A	B	AvB
1	1	1
1	0	1
0	1	1
0	0	0

Boolean logic - NOT

A	~A
1	0
0	1

Boolean logic uses

- Data analysis
- Searching the literature for key terms
- Developing critical thought

Question

- How do the trends in life expectancy for Australians compare with those living in Ireland?
  - We need boolean logic!
    - country == "Australia" or country == "Ireland"

Question

- How do the trends in life expectancy for Australians compare with those living in Ireland and Brazil?

Some further boolean logic questions!

- OR:
  - lifeExp > 50 or year == 1952
  - country == “Australia” or year == 2002
- AND:
  - pop > 5000000 and year == 1952
  - pop > 5000000 and lifeExp > 80
- NOT:
  - year != 1992
  - country != “Ireland”

Question

- What is the correlation between GDP per capita and life expectancy?
- How do these values compare with a linear regression between GDP per capita and life expectancy?

Question

- In 2007, what was the correlation between GDP per capita and life expectancy?

Question

- Use descriptive statistics to see how the mean life expectancy varies by each continent in 2007
- Conduct a linear regression between life expectancy (dependent variable) and continent (factor)
- Compare the descriptives with the linear regression output. Notice any similarities?
  - Note the reference levels can be changed for continent by navigating to "Reference Levels"

Question

- How much additional variance in life expectancy does GDP per capita explain over and above continent

Questions?

- You can ask questions any time on the discussion forums.
  - Jamovi
  - Tutorial